

1. Introduction to Determinants

A **determinant** is a scalar value that is calculated from a **square matrix**.

The determinant of a matrix A is written as:

$$\det(A)$$

or:

$$|A|$$

The determinant gives important information about a matrix, such as:

- Whether a matrix has an inverse
- Whether a system of equations has a unique solution
- The scaling factor of transformations in geometry

A determinant can only be calculated for a **square matrix**.

Examples of square matrices:

$$2 \times 2, \quad 3 \times 3, \quad 4 \times 4$$

Non-square matrices such as 2×3 do not have determinants.

2. Determinant of a 2×2 Matrix

For a matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

the determinant is:

$$\det(A) = ad - bc$$

Rule:

Multiply the main diagonal and subtract the product of the other diagonal.

Diagram:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Main diagonal:

$$ad$$

Other diagonal:

$$bc$$

Therefore:

$$\det(A) = ad - bc$$

3. Example: Determinant of a 2×2 Matrix

Find the determinant of:

$$A = \begin{bmatrix} 3 & 5 \\ 2 & 7 \end{bmatrix}$$

Using:

$$\det(A) = ad - bc$$

Substitute:

$$\det(A) = (3)(7) - (5)(2)$$

Calculate:

$$= 21 - 10$$

Therefore:

$$\boxed{\det(A) = 11}$$

4. Determinant of a 3×3 Matrix

For:

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

the determinant can be found using **cofactor expansion**.

Expanding along the first row:

$$\det(A) = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

This becomes:

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

5. Cofactor Expansion Method

The determinant can be found by expanding along any row or column.

For each element:

1. Select an element.
2. Remove its row and column.
3. Find the determinant of the remaining smaller matrix.
4. Apply the correct sign.

The sign pattern is:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

6. Example: Determinant of a 3×3 Matrix

Find:

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Expand along the first row:

$$\det(A) = 2 \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 3 \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 1 \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix}$$

Calculate each minor:

First minor:

$$\begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} = (5)(9) - (6)(8) \\ = 45 - 48 = -3$$

Second minor:

$$\begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} = (4)(9) - (6)(7) \\ = 36 - 42 = -6$$

Third minor:

$$\begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} = (4)(8) - (5)(7) \\ = 32 - 35 = -3$$

Substitute:

$$\det(A) = 2(-3) - 3(-6) + 1(-3) \\ = -6 + 18 - 3$$

Therefore:

$\det(A) = 9$

7. Determinant of a 4×4 Matrix

A 4×4 determinant is calculated by reducing it into smaller determinants.

Consider:

$$A = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix}$$

Expanding along the first row:

$$\det(A) = aC_{11} + bC_{12} + cC_{13} + dC_{14}$$

where each cofactor C_{ij} is a 3×3 determinant.

The expansion becomes:

$$\det(A) = a \begin{vmatrix} f & g & h \\ j & k & l \\ n & o & p \end{vmatrix} - b \begin{vmatrix} e & g & h \\ i & k & l \\ m & o & p \end{vmatrix} + c \begin{vmatrix} e & f & h \\ i & j & l \\ m & n & p \end{vmatrix} - d \begin{vmatrix} e & f & g \\ i & j & k \\ m & n & o \end{vmatrix}$$

Each 3×3 determinant is then solved using the 3×3 determinant formula.

8. Example: Finding the Determinant of a 4×4 Matrix

Find:

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 4 & 5 & 6 & 0 \\ 0 & 7 & 8 & 9 \\ 2 & 0 & 1 & 5 \end{bmatrix}$$

Expand along the first row:

$$\det(A) = 1C_{11} - 2C_{12} + 0C_{13} + 3C_{14}$$

The third term disappears because it contains zero.

First cofactor

Remove row 1 and column 1:

$$C_{11} = \begin{vmatrix} 5 & 6 & 0 \\ 7 & 8 & 9 \\ 0 & 1 & 5 \end{vmatrix}$$

Using the 3×3 formula:

$$= 5 \begin{vmatrix} 8 & 9 \\ 1 & 5 \end{vmatrix} - 6 \begin{vmatrix} 7 & 9 \\ 0 & 5 \end{vmatrix}$$

$$+0 \begin{vmatrix} 7 & 8 \\ 0 & 1 \end{vmatrix}$$

Calculate:

$$\begin{aligned} &= 5(40 - 9) - 6(35 - 0) \\ &= 5(31) - 6(35) \\ &= 155 - 210 = -55 \end{aligned}$$

Second cofactor

Remove row 1 and column 2:

$$C_{12} = \begin{vmatrix} 4 & 6 & 0 \\ 0 & 8 & 9 \\ 2 & 1 & 5 \end{vmatrix}$$

Calculate:

$$\begin{aligned} &= 4 \begin{vmatrix} 8 & 9 \\ 1 & 5 \end{vmatrix} - 6 \begin{vmatrix} 0 & 9 \\ 2 & 5 \end{vmatrix} \\ &= 4(40 - 9) - 6(0 - 18) \\ &= 4(31) - 6(-18) \\ &= 124 + 108 = 232 \end{aligned}$$

Third required cofactor

Remove row 1 and column 4:

$$C_{14} = \begin{vmatrix} 4 & 5 & 6 \\ 0 & 7 & 8 \\ 2 & 0 & 1 \end{vmatrix}$$

Calculate:

$$\begin{aligned} &= 4 \begin{vmatrix} 7 & 8 \\ 0 & 1 \end{vmatrix} - 5 \begin{vmatrix} 0 & 8 \\ 2 & 1 \end{vmatrix} + 6 \begin{vmatrix} 0 & 7 \\ 2 & 0 \end{vmatrix} \\ &= 4(7) - 5(-16) + 6(-14) \\ &= 28 + 80 - 84 \end{aligned}$$

$$= 24$$

Now substitute:

$$\begin{aligned}\det(A) &= 1(-55) - 2(232) + 3(24) \\ &= -55 - 464 + 72\end{aligned}$$

Therefore:

$$\boxed{\det(A) = -447}$$

9. Important Properties of Determinants

1. Determinant of Identity Matrix

$$\det(I) = 1$$

2. Determinant of Product

$$\det(AB) = \det(A) \det(B)$$

3. Determinant of Transpose

$$\det(A^T) = \det(A)$$

4. Swapping Two Rows

If two rows are exchanged:

$$\det(\text{new matrix}) = -\det(A)$$

5. Zero Determinant

If:

$$\det(A) = 0$$

then:

- The matrix is singular.
 - The inverse does not exist.
 - Rows or columns are linearly dependent.
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10. Applications of Determinants

Determinants are used for:

- Finding matrix inverses
 - Solving systems of equations using Cramer's rule
 - Checking linear independence
 - Calculating areas and volumes
 - Transformations in geometry
 - Eigenvalue calculations
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Summary

Matrix Size	Determinant Method
1×1	The single element
2×2	$ad - bc$
3×3	Cofactor expansion or Sarrus' rule
4×4 and larger	Cofactor expansion into smaller determinants

The determinant is the foundation for finding **adjugates and inverses**, because the inverse formula depends on:

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)}$$